

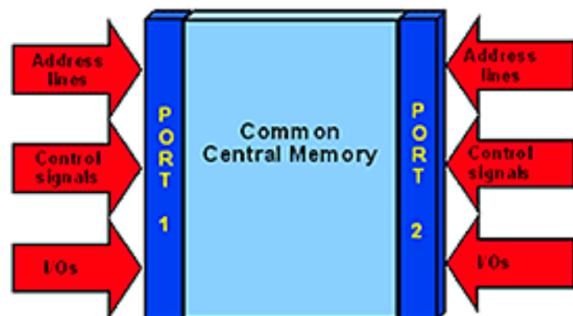
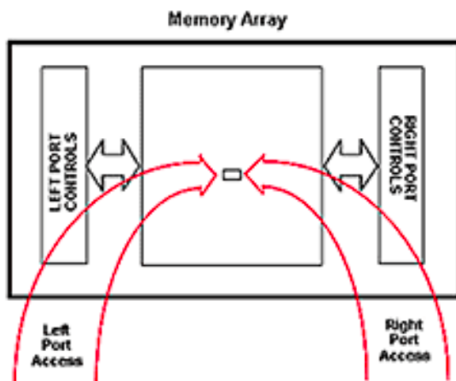
CAD-HW04-exe4

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برای طراحی یک RAM که دارای دو سر پورت برای کنترل حافظه است (Dual-Port Memory) نیاز به چنین ورودی ها و خروجی هایی داریم:

- Generic: Data_Width, Addr_Width (Default = 8)
- Inputs: CLK, WE1, RE1, Addr1, WE2, RE2, Addr2
- InOut: Data1, Data2



کد برنامه :

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_unsigned.all;
use IEEE.numeric_std.all;

entity exe4 is
    generic(
        Data_Width, Addr_Width : integer := 8
    );
    port(
        CLK, WE1, RE1, WE2, RE2: in std_logic;
        Addr1, Addr2: in std_logic_vector (Addr_Width - 1 downto 0);
        Data1, Data2: inout std_logic_vector (Data_Width-1 downto 0)
    );
end exe4;

architecture exe4 of exe4 is
    type RAM_TYPE is array(0 to 2**Addr_Width - 1) of std_logic_vector (Data_Width - 1 downto 0);
    signal RAM : RAM_TYPE;
begin
    process(CLK)
    begin
        if (CLK'event and CLK = '1') then
            if (WE1 = '1') then
                RAM(to_integer(unsigned(Addr1))) <= Data1;
            end if;
            if (WE2 = '1') then
                RAM(to_integer(unsigned(Addr2))) <= Data2;
            end if;
        end if;
    end process;
    Data1 <= RAM(to_integer(unsigned(Addr1))) when (RE1 = '1')
        else (others => 'Z');
    Data2 <= RAM(to_integer(unsigned(Addr2))) when (RE2 = '1')
        else (others => 'Z');
end exe4;
```

سپس کد را compile کرده و فایل test bench برای شبیه سازی می سازیم. و به آن کد زیر را اضافه می کنیم:

```
CLK <= '0', '1' after 5ns, '0' after 10ns, '1' after 15ns, '0' after 20ns, '1' after 25ns, '0' after 30ns, '1' after 35ns,
'0' after 40ns, '1' after 45ns, '0' after 50ns, '1' after 55ns, '0' after 60ns, '1' after 65ns, '0' after 70ns;
WE1 <= '0', '1' after 10ns, '0' after 20ns, '1' after 30ns, '0' after 40ns, '1' after 50ns, '0' after 60ns, '1' after 70ns;
RE1 <= '0', '1' after 15ns, '0' after 30ns, '1' after 45ns, '0' after 60ns;
WE2 <= '0', '1' after 10ns, '0' after 20ns, '1' after 30ns, '0' after 40ns, '1' after 50ns, '0' after 60ns, '1' after 70ns;
RE2 <= '0', '1' after 15ns, '0' after 30ns, '1' after 45ns, '0' after 60ns;
Addr1 <= "00000000";
Addr2 <= "00000001";
Data1 <= "01010101", "11001100" after 32ns;
Data2 <= "00111100", "01011101" after 32ns;
```

بعد از compile کردن فایل test bench آن را شبیه سازی می کنیم. برای این کار فایل test bench را excute می کنیم. و سپس برنامه را run می کنیم.

نتیجه ی شبیه سازی:

